

Numerical Method

CSIT — 3rd Semester — 2079 — Answer Sheet
Subject Code: CSC212

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. How secant method can approximate the root of a non-linear equation? Explain with necessary derivation. Estimate a real root of following equation using secant method. Assume error precision of 0.01. $x^3 + 2x - \cos(x) = 4$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. How spline interpolation differs with the Lagrange's interpolation? Estimate the value of $f(0)$ and $f(4)$ using cubic spline interpolation from the following data. $x-1135f(x)-10-21486$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What is pivoting? Why is it necessary? Write an algorithm and program to solve the set of n linear equations using Gaussian elimination method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Calculate a real root of the following function using bisection method correct up to 3 significant figures. $x^2 - e^{x-1} = 3$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is fixed point iteration method? How can it converge to the root of a non-linear equation? Also explain the diverging cases with suitable examples.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Write down program for solving ordinary differential equation using Heun's method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Fit the quadratic function for the data given below using least square method.

x	f(x)
1.01	2.745
1.02	2.88
1.03	3.11
1.04	2.15
1.05	1.9

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Estimate the integral value of following function from $x = 1.2$ to $x = 2.4$ using Simpson's 1/3 rule.

x	f(x)
1.2	1.532
1.4	2.253
1.6	1.84
1.8	3.25
2.0	6.77
2.2	2.38
2.4	9.413

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is Gaussian integration formula? Evaluate the following integration using Gaussian integration three ordinate formula.

$$\int_0^1 (\sin x)/x \, dx$$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Solve the following set of equations using Gauss Seidel method.

$$\begin{aligned} x + 2y + 3z &= 46 \\ x + 4y + 5z &= 165 \\ x + 2y + 3z &= 12 \end{aligned}$$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Solve the following differential equation for $0 \leq x \leq 1$, taking $h = 0.5$ using Runge Kutta 4th order method.

$$y'(x) + y = 3x, \text{ with } y(0) = 2$$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Solve the Poisson's equation $\nabla^2 f = 3x^2y$ over the square domain $0 \leq x \leq 3, 0 \leq y \leq 3$ with $f = 0$ on the boundary and $h = 1$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.